Website Analytics User Guide

A practical guide to using the GA4 desktop dashboard

Last changed: July 23, 2026

All screenshots in this guide are actual application screenshots that have been cropped and sanitized. Private property names, property IDs, OAuth values, personal paths, and location details are hidden.

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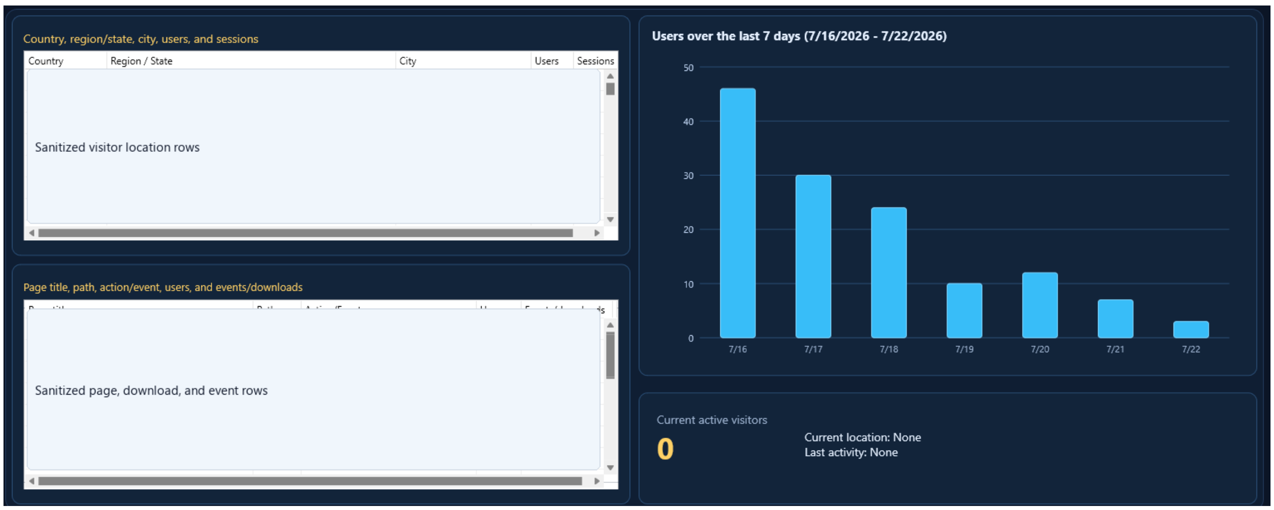
1. What This Application Is Doing

Website Analytics is a Delphi FMX desktop dashboard for Google Analytics 4. Its purpose is to answer the everyday questions a site owner actually asks: who visited, where they came from, what pages or downloads they used, what the recent trend looks like, and whether anyone is active right now. It does not try to reproduce every GA4 screen. It pulls selected GA4 data through the Google Analytics Data API, keeps the report rows in memory, formats the results, and displays them in a cleaner desktop layout.

The application stores settings locally, but analytics report results are intentionally memory-only. When the application closes, the fetched GA4 report rows are discarded. This keeps the program simple, portable, and suitable for open-source use without creating a second analytics database.



Actual dashboard screenshot, cropped/sanitized to hide private row data.

2. Important Terms

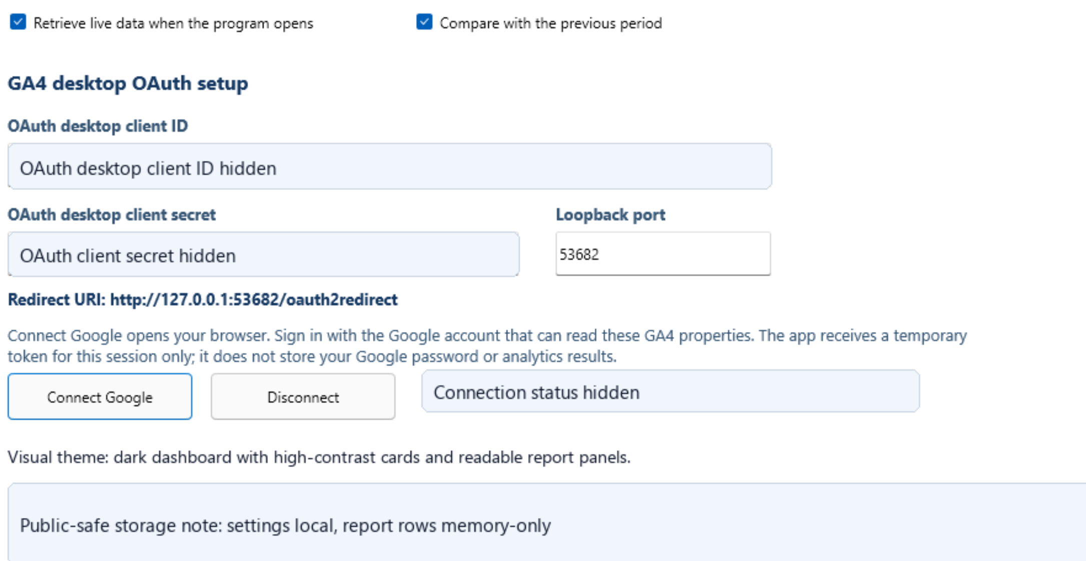
Term	Meaning in this application
Website property	A saved website definition that contains a display name and a numeric GA4 property ID.
GA4 property ID	The numeric property identifier used by the GA4 Data API. It is not the G- measurement ID.
Date range	The selected historical reporting window, such as Today, Last 7 days, Last 28 days, or Last 90 days.
Realtime data	GA4 data from the realtime endpoint, generally representing current or recent activity.
Standard report data	Historical GA4 report data returned by runReport requests for a selected date range.

Memory-only rows	Fetched analytics rows that are shown on screen but not saved to the SQLite database.

3. First-Time Setup

First-time setup connects your copy of the desktop app to your own Google account and your own GA4 properties. The app does not include shared Google credentials. Each user should create and enter their own OAuth desktop client values.

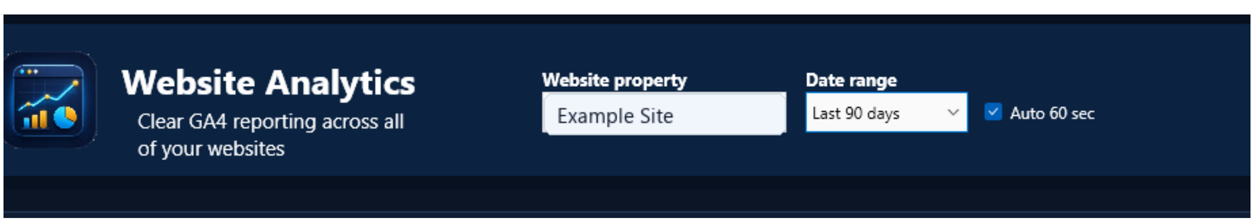
1. Confirm that GA4 is installed and collecting data on the website.
2. Find the numeric GA4 property ID in Google Analytics Admin.
3. Create a Google Cloud Desktop OAuth client for this app.
4. Enable the Google Analytics Data API in the same Google Cloud project.
5. Open Settings in the app and enter the OAuth desktop client ID and client secret.
6. Confirm that the redirect URI shown in Settings matches the local loopback redirect used by the desktop OAuth client.
7. Click Connect Google, complete the browser sign-in, return to the app, and click Update.



Actual Settings screen section with OAuth values redacted.

4. The Main Dashboard

The dashboard is organized for glanceable use. The top hero area contains the app identity, website selector, date range selector, auto-update setting, current connection status, and the Update, Settings, and Help buttons.

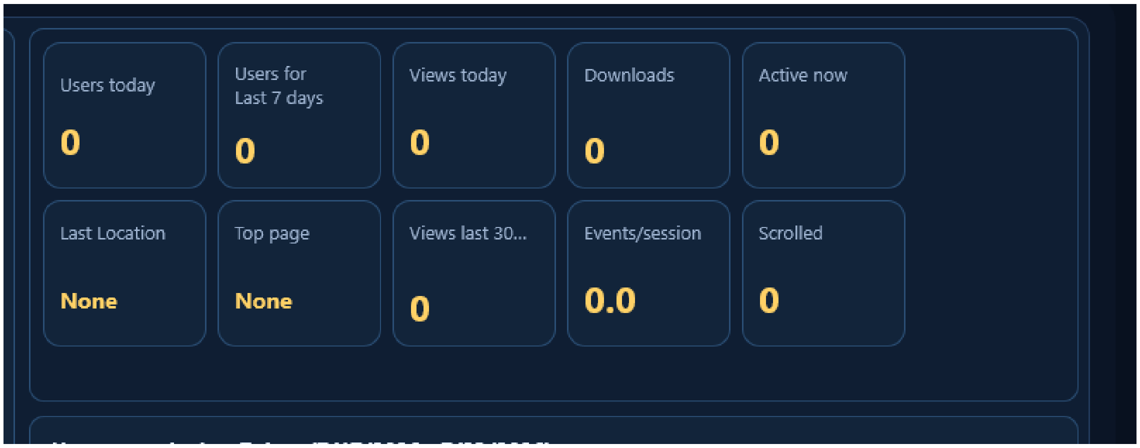


Actual hero/control area with the selected property value redacted.

Below the hero, the workspace is divided into practical report areas. The left side focuses on tabular lookup: visitor locations and pages/downloads/actions. The right side focuses on visual trend and realtime status.

5. Summary Cards

The card group gives quick values without forcing you to read a grid. These cards should be used as first-glance indicators. If something looks interesting, the grid or chart below the cards gives more context.



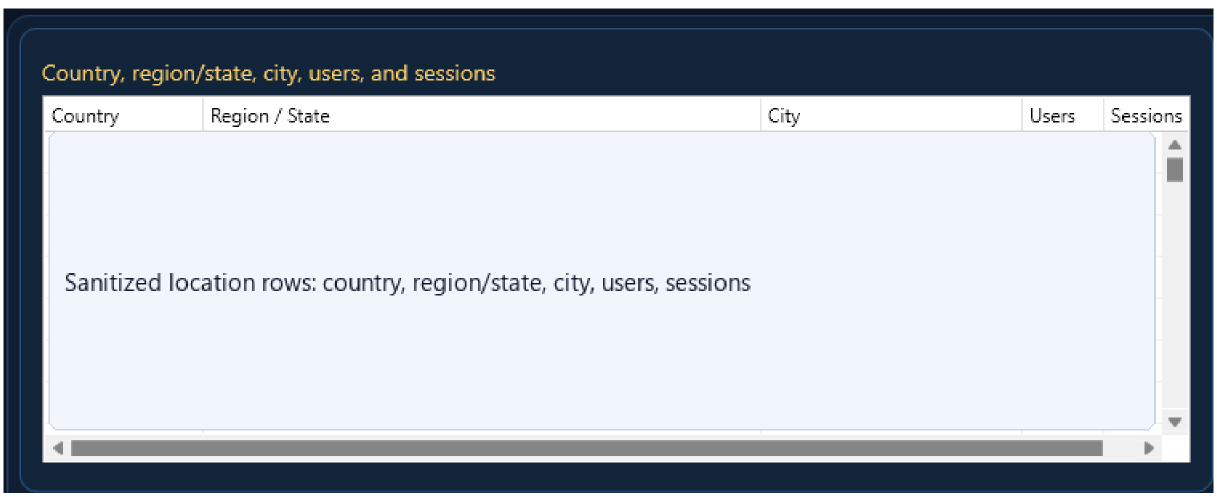
Actual summary-card area from the application.

Card	What it tells you	Notes
Users today	Users reported for the current day.	Uses the current day rather than the selected historical range.
Users for selected range	Users for the selected date range.	Should agree with the users chart total for the same reporting scope.
Views today	Page/screen views for the current day.	Useful for quick same-day activity checks.
Downloads	Download events in the selected range.	Depends on the website emitting GA4 download events.
Active now	Current realtime active users.	Realtime data can update differently from standard report data.
Last Location	Latest returned realtime location.	Shown only when GA4 returns usable city/country or city/state data.
Top page	Most-used page for the selected	Home page is shown when the GA4

	range.	path is /.
Views last 30 min	Realtime views during the recent realtime window.	Uses realtime reporting.
Events/session	Average events per session.	Helps indicate whether sessions are shallow or active.
Scrolled	Users who triggered the scroll metric/event.	Useful as a simple engagement signal.

6. Visitor Locations

The location grid answers where visitors came from. It groups by country, region/state, and city, then shows users and sessions. Rows are sorted alphabetically so a country, state, or city is easier to find than in GA4's default scrolling lists.

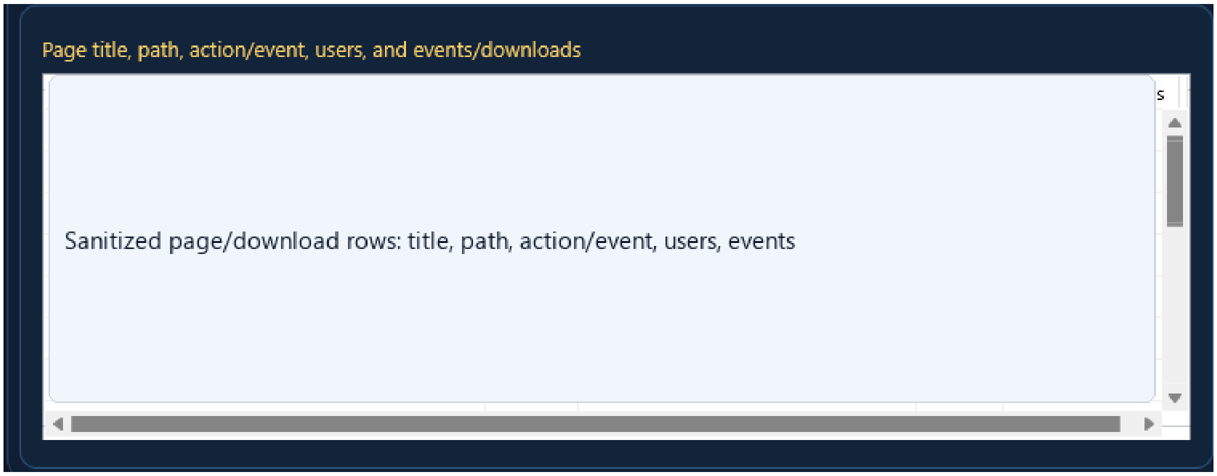


Actual location grid with visitor rows redacted.

- Country is the broadest location grouping.
- Region / State is a state, province, county, or other regional value when GA4 supplies it.
- City is shown when GA4 can resolve it; otherwise GA4 may return (not set).
- Users counts people; sessions counts visits/sessions. They are related but not the same number.
- If a row says (not set), that is a GA4 result, not a dashboard formatting bug.

7. Pages, Downloads, and Actions

The pages/downloads area answers what visitors used. Instead of burying this information inside GA4 event tables, the dashboard combines page title, path, action/event, users, and event/download counts into one practical view.

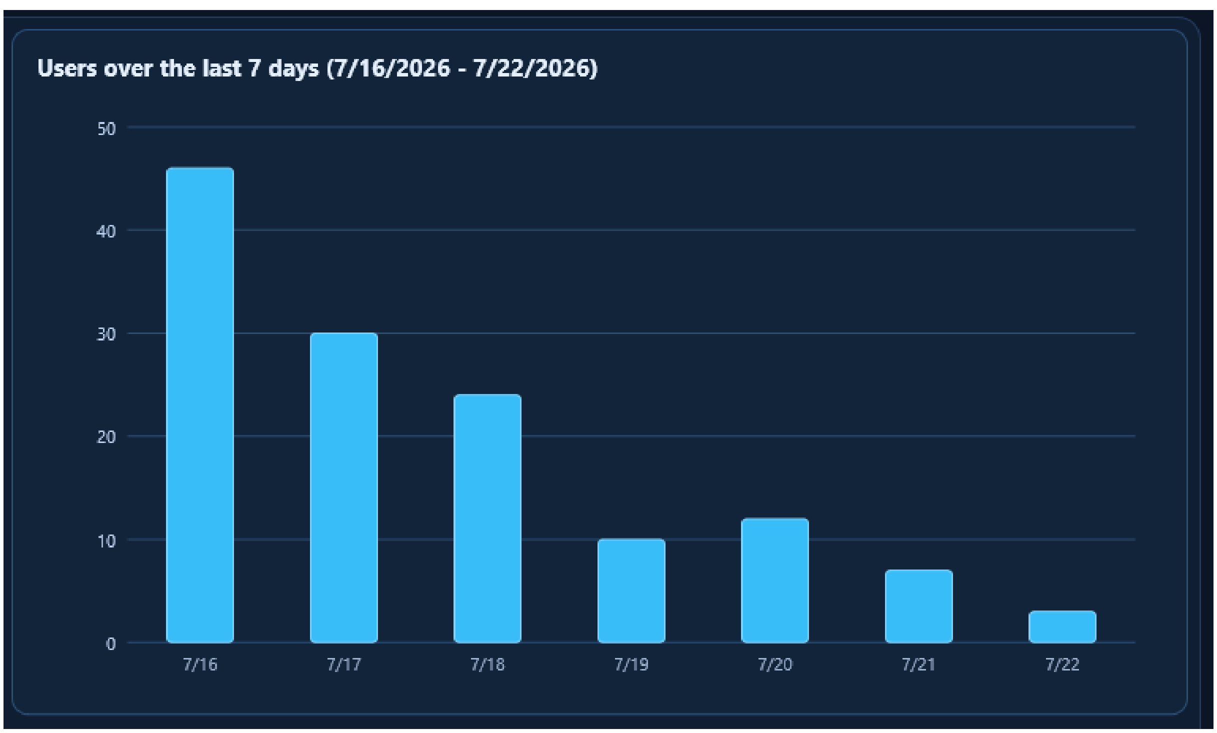


Actual pages/downloads area with row details redacted.

- Page title is the browser/page title reported to GA4.
- Path is the URL path, such as / or /downloads.html.
- Action/Event distinguishes page views, file downloads, clicks, scrolls, and other GA4 events.
- Users shows how many users were associated with that row.
- Events/downloads shows how many event occurrences were returned for that row.

8. Users Graph

The users graph is intentionally simple. Its job is to show whether traffic rose, fell, spiked, or stayed flat over the selected date range. The current layout uses a bar graph because it is easier to read than a smooth line when values are daily counts.



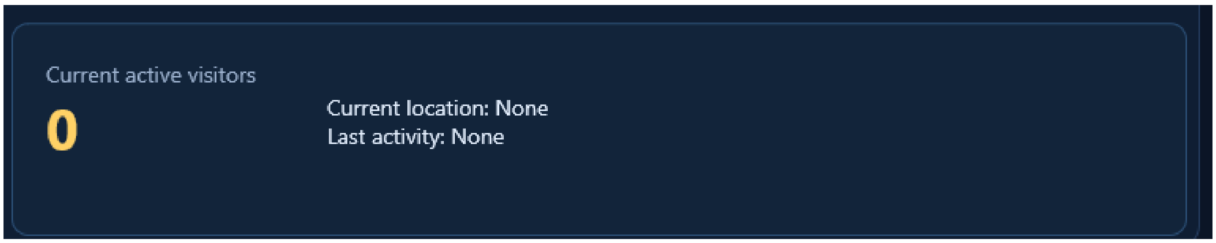
Actual users graph from the dashboard.

- The title shows the selected range and exact date span.

- The left scale uses whole numbers and starts at zero.
- Each bar represents the reporting unit shown on the chart, usually a day.
- Hovering over a bar shows the exact value in the compact tooltip.

9. Realtime Activity

The realtime panel answers what is happening right now. This information comes from GA4 realtime reporting and should be treated as a live indicator, not as a permanent historical record. When the selected property changes, realtime location text should reset so stale locations are not carried between sites.



Actual realtime panel with no private location shown.

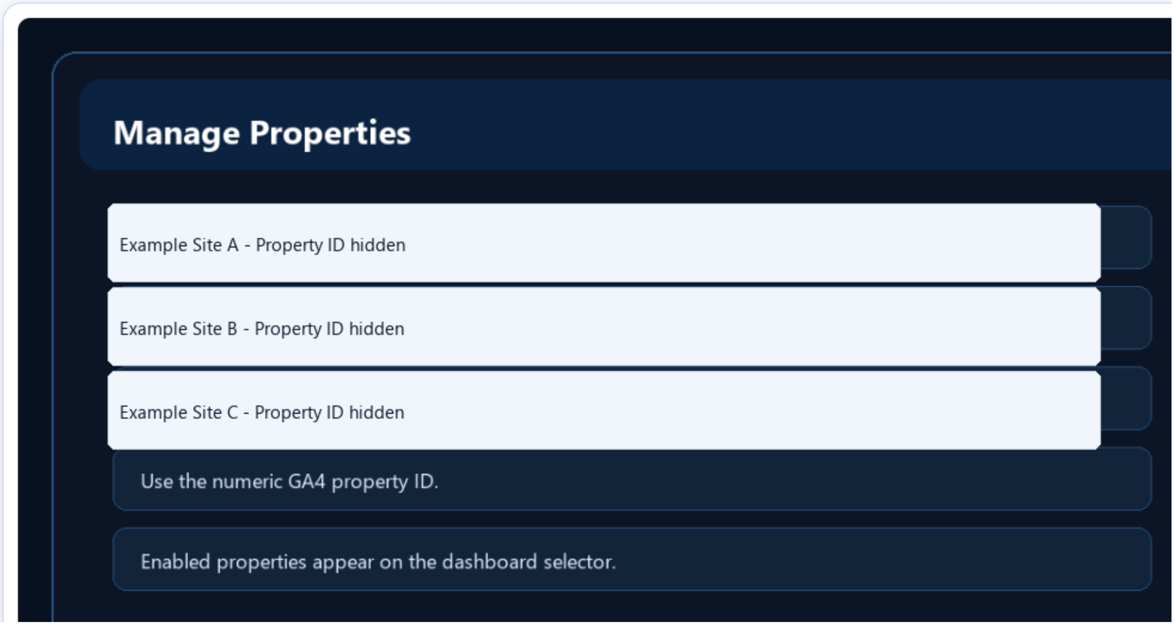
10. Auto Update and Long-Running Use

The app is intended to be left open while you work. With Auto 60 sec enabled, it refreshes automatically about once per minute. Manual Update is still available when you want an immediate refresh. If the access token expires, the app attempts to refresh it silently using the encrypted refresh token.

- Leave the app open for an always-visible site dashboard.
- Use the status bar to confirm the last update or see an error message.
- If Google sign-in appears again, the saved refresh token may have been revoked, removed, or tied to another Windows account.
- Realtime and standard reports can differ briefly because GA4 updates those data surfaces differently.

11. Managing Properties

Properties are the websites the app can query. A property definition normally includes a display name, URL, numeric GA4 property ID, enabled state, and ordering information. Only enabled properties appear in normal dashboard selection.



Actual Manage Properties screen with property names and IDs redacted.

- Use the numeric GA4 property ID from GA4 Admin.
- Do not enter the G- measurement ID.
- Do not enter the Google Cloud project number.
- For open-source copies, each user should add their own properties.

12. Privacy and Storage

The dashboard is intentionally conservative about storage. Settings are stored so the app can remember how you use it. Report results are not stored. The saved Google refresh token is encrypted using Windows DPAPI so it is tied to the current Windows user profile.

Stored	Where	Why
Dashboard settings	Portable SQLite database	Remember defaults such as date range and selected website.
Website/property definitions	Portable SQLite database	Let the app query multiple websites.
OAuth client ID/secret	Portable SQLite database	Allow Google token exchange for the desktop OAuth client.
Refresh token	SQLite, encrypted with DPAPI	Allow silent reconnect on startup.
GA4 report rows	Memory only	Avoid creating a separate analytics database.

13. Troubleshooting

Symptom	Likely cause	What to check
Google says the app is not verified	OAuth consent app is still in testing	Add yourself as a test user or complete Google verification before broad public use.
403 permission error	Wrong property ID or wrong Google account	Confirm the numeric GA4 property ID and GA4 Admin access for the signed-in account.
No rows returned	No GA4 data for that range/property	Try a wider date range and compare with GA4.
Downloads are zero	No download event reported	Confirm the website sends file_download or equivalent GA4 events.
Realtime shows location but grid does not	Realtime and standard GA4 reports update differently	Refresh later or compare current-day standard reports.
Login appears again	Refresh token missing/revoked/unusable	Open Settings and reconnect Google.

14. How to Interpret the Numbers

The dashboard intentionally separates similar-looking numbers because GA4 metrics do not all mean the same thing. Users, sessions, views, events, downloads, realtime active users, and realtime views answer different questions. The most common mistake is expecting every number to move together. A single person can create multiple sessions, view multiple pages, trigger several events, and still count as one user.

Metric	Plain-language meaning	How to use it
Users	People or browser/device identities GA4 counted.	Use for overall audience size.
Sessions	Visits or activity periods.	Use for visit volume. Sessions can exceed users.
Views	Page/screen views.	Use for page consumption and site activity.

Events	Tracked actions GA4 received.	Use for interactions such as clicks, downloads, scrolls, and engagement.
Downloads	Download-related events.	Use to judge whether users are getting files or documents.
Events/session	Average event depth per session.	Use as a compact engagement indicator.
Scrolled	Users who triggered scroll tracking.	Use to see if visitors are moving beyond the first screen.
Active now	Realtime active users.	Use as a live pulse, not a historical total.

For a quiet website, daily numbers can be small. That is not a failure. The value of the dashboard is that it makes small numbers readable, keeps the complete rows visible, and avoids GA4's habit of hiding most values behind limited tables and side-scrolling panels.

15. Date Ranges and All-Websites Mode

The selected date range controls most historical panels. Today is useful for checking whether the site is alive right now. Last 7 days is good for a short trend. Last 28 days gives a month-like operational view. Last 90 days is better for slow-moving sites where one week may not show enough activity.

All websites mode combines enabled properties. This is useful for a broad operational view, but it can also make labels less specific because the same page path or event name may exist on more than one website. For investigation, switch back to one property at a time. Individual property review is usually clearer when you want to know exactly which website produced a row.

- Use Today to verify new tracking changes or a current visitor.
- Use Last 7 days to see whether traffic changed this week.
- Use Last 28 or Last 90 days to reduce noise on low-traffic sites.
- Use All websites for a quick roll-up, then switch to individual properties for detail.

16. Google OAuth Walkthrough

Google OAuth is the part of the setup that feels the least obvious at first. The important idea is that the desktop app does not use your email address as a client ID. The app needs a Google Cloud Desktop OAuth client. That client gives you a client ID and often a client secret. Those two values identify the desktop app during Google's authorization flow.

8. Open Google Cloud Console and select or create a project for this desktop dashboard.
9. Enable the Google Analytics Data API in that project.
10. Configure the OAuth consent screen. While testing, add your Google account as a test user.
11. Create an OAuth Client ID with application type Desktop app.

12. Copy the generated client ID and client secret into Settings.
13. Use Connect Google from Settings. Google opens in the browser, you approve access, and the browser returns to the app's local redirect URI.
14. After success, the refresh token is encrypted locally and future launches should reconnect silently.

If you later publish the project publicly, each user should create their own OAuth client. Do not publish your client secret or local database. For wide public distribution, Google app verification may be required so users do not see unverified-app warnings.

17. Public/Open-Source User Notes

A public copy of this project should not include the developer's SQLite database, OAuth secret, refresh token, property IDs, or personal screenshots. Each user cloning the repository should create their own Google Cloud OAuth desktop client, add their own GA4 property IDs, and let the app create a local settings database at startup.

Public repo item	Should it be included?	Reason
Source code and forms	Yes	Needed to build and inspect the app.
Help and guides	Yes	Needed for users and developers.
Schema SQL	Yes	Allows fresh database creation.
Real .sqlite3 database	No	Can contain settings, property IDs, and tokens.
OAuth client secret	No	Credential material.
Refresh token	No	Account authorization material.
Real screenshots with private data	No	Can reveal property IDs, locations, account details, and usage.

18. Daily Operating Checklist

- Open the app and confirm the status shows Connected.
- Choose the property you want to review.
- Choose the date range that matches the question you are asking.
- Check the cards first for the quick story.
- Check the users graph for spikes or drop-offs.
- Check locations to see where visitors came from.
- Check pages/downloads to see what they used.
- Check realtime if you are actively testing or watching a current visitor.

- Leave Auto 60 sec enabled if you want the dashboard to run continuously.